

WebGeoInfer: Structure-Free Multi-Stage Framework for Geolocation Inference from Exposed Device Web Interfaces

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Abstract

While the web interfaces of remotely managed devices offer convenience, their unstructured content can inadvertently leak geographic locations, posing a significant security risk. We aim to assess the feasibility of automatically exploiting this leakage, serving as a clear warning to cybersecurity regulators. To this end, we propose WebGeoInfer, a framework that does not rely on page structure. It extracts clues through page clustering and differential analysis to overcome the challenge of information heterogeneity. It also leverages search engines and large language models to augment sparse clues and infer precise coordinates, addressing the challenge of information sparsity. In large-scale experiments, WebGeoInfer successfully located 5,435 devices across 94 countries and 2,056 cities, achieving accuracy rates as high as 96.96% at the country level, 88.05% at the city level, and 79.70% at the street level. These findings provide the first conclusive evidence of the reality and scale of this threat. Furthermore, our analysis offers new insights and mitigation strategies for affected devices, establishing a key benchmark for future security research.

CCS Concepts

• **Security and privacy** → *Network security*; • **Information systems** → **Web searching and information discovery**.

Keywords

Cybersecurity, Geolocation Inference, Large Language Models, Attack Surface Analysis

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1 Introduction

The use of remote management devices is crucial in modern infrastructure operations, offering administrators unprecedented convenience to efficiently monitor and control geographically dispersed assets such as power grid nodes, water treatment facilities, or building automation systems [2, 21, 29, 39]. While web interfaces offer significant convenience, their public exposure raises a critical security concern: Does this convenience compromise system security? These highly customizable pages contain vast amounts of information, leading us to our core security question: ***Does this information contain sensitive geolocation data? Can attackers automatically extract accurate location information from these unstructured and diverse web pages?*** Answering this question is crucial, as it directly impacts our ability to provide effective warnings for cyberspace asset regulation and uncovers a potential, widely overlooked attack surface [24, 31, 36, 42].

To answer our core security question, its feasibility hinges on two key techniques: **web content analysis** and **geolocation inference**. In the domain of web content analysis, existing research has primarily focused on extracting information from structured or semi-structured web pages, for example, using wrappers based on DOM structures or semantic tags [5, 9, 17, 28]. In the domain of geolocation inference, both passive and active methods have their specific application scopes. Passive methods, like DNS analysis or Whois queries [8, 27], rely on relatively explicit and complete data, while active methods perform localization by measuring external signals like network latency [23, 26].

However, these existing methods offer limited contributions to our problem, as they are ill-equipped to handle two fundamental challenges we face: **First, The Information Heterogeneity Challenge:** The sheer diversity in web interfaces across thousands of device types renders rule-based or structure-dependent extraction methods [9] almost useless, as there is no unified design standard. **Second, The Information Sparsity Challenge:** Unlike the explicit 'data' (e.g., country codes in hostnames) used by traditional

passive methods [27], the clues on device pages are often ambiguous, fragmented 'clues' (e.g., "Main St. Branch"), which existing methods are not designed to handle through the necessary deep contextual inference [11].

To address this challenge, we designed WebGeoInfer, a framework specifically aimed at answering our core security question. To overcome heterogeneity, we introduced a "web-structure-free" differential text extraction module. Through page clustering and intra-cluster analysis, it automatically isolates critical, administrator-customized clues from noisy background information without requiring rules for any specific device. To test the feasibility of overcoming sparsity, we constructed a multi-stage information enhancement pipeline. This process integrates search engines and Large Language Models (LLMs), leveraging their powerful contextual understanding and reasoning to piece together fragmented clues into complete geographic coordinates.

Our experimental results provide a clear answer: **it is entirely feasible**. WebGeoInfer successfully located 5,435 devices across 94 countries and 2,056 cities. This achievement not only validates our technical approach but, more importantly, demonstrates that the automated extraction of geolocations from web pages represents a legitimate and scalable attack vector.

In summary, the contributions of this research can be summarized as the following three points:

- **Web-structure-free differential text extraction.** By leveraging page clustering and intra-cluster analysis, this method solves the information heterogeneity problem caused by diverse device interfaces.
- **A multi-stage inference framework integrating search engines and LLMs.** Through progressive clue refinement and reasoning, this framework overcomes the information sparsity challenge of geographic clues.
- **The first large-scale empirical evaluation of this attack surface.** This evaluation achieves 79.7% street-level localization accuracy across thousands of real-world devices, providing the first quantifiable benchmark for this overlooked threat.

2 Related Work

Our objective is to analyze the feasibility of extracting geolocation information at a large scale from non-standardized web pages exposed by network devices. Therefore, we analyze related work from two perspectives: Web content extraction and device geolocation methods.

2.1 Web Content Extraction Methods

Current web content extraction methods generally rely on some form of "structure" or "rules" [1, 3, 35, 37, 41]. For instance, some research attempts to use Named Entity Recognition (NER) [4] or semantic analysis [9, 32] to find place names. While these methods perform reasonably well on standardized text (such as news articles), they fail completely when faced with thousands of device web interfaces with vastly different structures. Training a dedicated model or writing a set of extraction rules for each interface type is practically impossible, making them unsuitable for validating "at-scale" feasibility. Although Zilberman et al. [42] pointed out

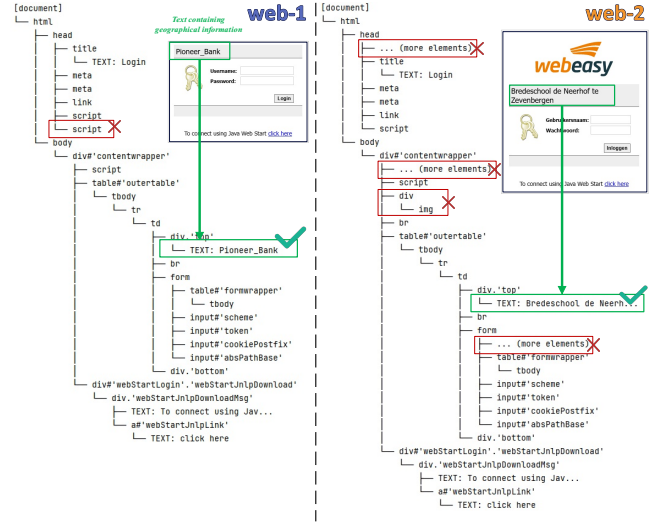


Figure 1: Comparing DOM tree differences across similar web pages.

the potential of extracting information from device web interfaces, they did not provide a systematic solution to overcome this extreme heterogeneity.

2.2 Device Geolocation Methods

Existing device geolocation methods fall into two categories: passive geolocation and active geolocation.

Existing passive geolocation inference methods rely on relatively explicit geographical clues [25, 30, 40]. For example, DNS analysis methods [10, 19, 27] attempt to find city or country codes within hostnames, while database query methods [8, 15, 20] directly match IP addresses. The commonality of these methods is their need for directly usable and relatively complete "data," rather than "clues" that require deep inference. They cannot handle the ambiguous, fragmented, and highly context-dependent sparse information we face, such as "R&D Center, Building A, 3rd Floor."

As for active probing methods [12, 33, 34], their approach is fundamentally different from ours. They measure latency by sending network packets, which is not only susceptible to network fluctuations and firewall policies but also unable to leverage the richer textual information exposed by the device pages themselves.

In summary, existing work either relies on fixed page structures, which cannot overcome heterogeneity, or on explicit geolocation data, which cannot overcome sparsity. There is a lack of methods that can simultaneously address the core challenges of page heterogeneity and information sparsity.

3 Motivation

The core motivation of our research is to provide a clear warning for cybersecurity regulation: automatically extracting geolocations from the web pages of exposed remote management devices is not merely a theoretical possibility but a practical and scalable threat. To issue this warning, we must first answer a critical feasibility

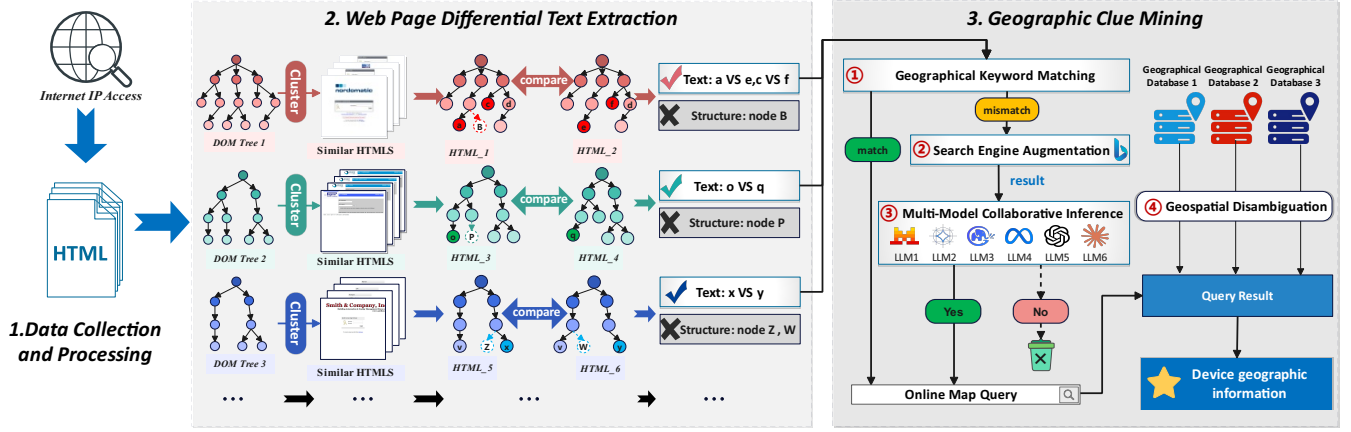


Figure 2: Overall framework. Including Data Collection and Processing, Web Page Differential Text Extraction and Geographic Clue Mining.

question with empirical evidence: can we systematically overcome the two core obstacles in the automated extraction process?

The Heterogeneity Obstacle: Device web pages lack design standards, resulting in vastly different structures. Customizing rules for each device is impractical.

Key 1: Can we find a universal method to identify geographic information without relying on specific page structures?

Our Approach: Ignore Structure, Focus on Differences. We observed that pages of similar devices are mostly identical, while customized information entered by administrators (including geolocations) exists precisely in the "differential" parts (Fig. 1). Therefore, by clustering pages and performing intra-cluster differential analysis, we can automatically isolate these high-value clues, thus proving that overcoming heterogeneity is feasible.

The Sparsity Obstacle: The extracted clues are often fragmented and ambiguous (e.g., building acronyms), offering low direct value.

Key 2: Can we reliably infer precise coordinates from these low-quality, sparse clues?

Our Approach: Multi-Stage Enhanced Inference. We designed a pipeline that emulates expert geographic inference strategies: first, we use search engines to "add context" to sparse clues; next, we leverage multiple LLMs for "collaborative reasoning" to enhance robustness [6, 7, 18]; finally, we perform "ambiguity resolution" using geographical databases. This process is designed to prove that we can recover complete information even from sparse clues.

4 Framework

To systematically address these obstacles, answer the key questions above. We propose a comprehensive framework called WebGeoInfer consisting of three core components: data collection and processing, web page differential text extraction, and geographic clue mining. Figure 2 presents an overview of the framework.

4.1 Data Collection and Processing

Our study systematically collects web page information of remote management devices from the public internet using a comprehensive methodology. Operating within a strict regulatory framework, we examine information exposed by these devices on the public internet. The data collection process encompasses the following essential steps:

Web Page Access: We use browsers to access combinations of IP addresses and ports to obtain web page information from remote management devices. This access method is consistent with how ordinary users access public network resources. For pages requiring authentication, we exit immediately without attempting any form of hacking or cracking, ensuring full compliance with legal and ethical standards.

Page Capture : Comprehensive information collection of target device pages, primarily including:

- HTML source code (Original HTML page information).
- Page screenshots (Used for intuitive analysis of subsequent data annotation).

Data Preprocessing: Utilizes the lxml and BeautifulSoup libraries for structured HTML processing, extracting and standardizing DOM trees to establish a robust foundation for subsequent clustering analysis. This preprocessing workflow ensures both data consistency and analytical integrity.

4.2 Web Page Differential Text Extraction

To extract geographic text information embedded within page differences, we present a method based on structural similarity clustering and differential text extraction. This approach comprises two key stages: device page clustering and differential text extraction (as illustrated in Appendix Algorithms 2 and 1).

4.2.1 Device Page Cluster Stage. We implement a Nilsimsa-based Locality-Sensitive Hashing (LSH) scheme to cluster pages according to their structural similarity. The system serializes each page's DOM tree, generating a 128-bit hash digest, with similarity quantified through Hamming distance. Through hierarchical aggregation, the

method clusters pages with similarity exceeding threshold θ , creating compact clusters that share a common structural skeleton—thus establishing a robust foundation for subsequent text alignment.

The clustering process encompasses four essential steps: DOM tree serialization, Nilsimsa hash generation, similarity matrix construction, and hierarchical clustering. The similarity threshold θ (default: 40) is calibrated through comprehensive experimental analysis.

4.2.2 Differential Text Extraction Stage. Within each cluster $C_k \in C$, we execute structural alignment through subtree isomorphism search to precisely identify node groups across pages.

For each aligned node group, the system directly compares text content at identical structural positions. When the algorithm detects textual variations at equivalent structural positions, it identifies these differences as differential markers that frequently reveal geographical location clues. This direct comparison methodology circumvents complex similarity calculations, thereby enhancing efficiency and facilitating the capture of subtle yet critical textual variations.

4.3 Geographic Clue Mining

This section presents our core framework for extracting geographic location intelligence from differential text segments. To systematically address the challenges in differential texts that contain non-geographic elements (device attributes, status indicators), or geographic information presented in abbreviated or implicit forms, we implemented a progressive four-stage mining process, including: Geographic Keyword Matching, Search Engine Information Augmentation, Multi-Model Collaborative Inference, and Geospatial Disambiguation with Database Assistance. Each stage builds upon the previous, creating an integrated pipeline that progressively refines geographical inference.

4.3.1 First Stage-Deterministic Geographic Keyword Matching. In some differential texts, explicit geographic clues like "Gilbert High School" or "Georgia Intercounty Convention Center" can be identified. To rapidly filter this high-value information at low cost, this stage employs a keyword matching method. We adapt and define a library of 65 keywords based on existing geospatial information standards (e.g., ISO 19112), covering categories such as buildings, roads, commercial institutions, and administrative divisions (Appendix Table 3). If a differential text matches these keywords, it is directly identified as a clue with strong geographic characteristics. This method acts as an efficient pre-processor, significantly reducing the computational overhead and false positives for the more complex model inference stages that follow.

4.3.2 Second Stage-Search Engine Information Augmentation. This stage specifically addresses incomplete geographic signifiers that resist direct identification, particularly abbreviations, acronyms, and context-dependent references. When deterministic matching fails, the system leverages the capabilities of commercial search engines for semantic expansion. This technique submits these low-entropy textual fragments as search queries, then applies natural language processing to extract and consolidate geographic context from the search results. This semantic enrichment process significantly enhances the information density of the original text, creating a more

information-symmetric and robust foundation for the subsequent model-based inference and effectively bridging the gap between abbreviated references and their full geographical significance.

4.3.3 Third Stage-Multi-Model Collaborative Inference. In this stage, we leverage the exceptional capability of LLMs in both explicit pattern recognition and implicit semantic relationship inference. We deploy a heterogeneous ensemble of open-source and proprietary models, which are carefully selected through a comprehensive evaluation of their geographical inference performance, domain-specific capabilities, and computational efficiency. To enhance precision, we design a systematic set of Chain-of-Thought (CoT) prompting protocols that guides models through a logical reasoning sequence: first, entity recognition to locate potential geographic names; next, contextual verification to validate accuracy; and finally, an inference chain to resolve ambiguities.

To resolve disagreements among models, we implement a weighted ensemble decision system. The final geographical inference G^* is determined by:

$$G^* = \arg \max_{G \in \mathcal{G}} \sum_{i=1}^n w_i \cdot c_i(G) \quad (1)$$

where G^* represents the final geographical inference result, w_i is the confidence weight assigned to model i based on its historical accuracy, and $c_i(G)$ is the confidence score that model i assigns to the geographical entity G .

This ensemble approach optimizes inference reliability by leveraging complementary strengths across diverse model architectures. The system then transforms extracted geographical entities into standardized coordinate representations through integration with geospatial mapping services.

4.3.4 Fourth stage-Geospatial Disambiguation with Database Assistance. Considering that online map queries often yield multiple candidate locations, this final stage aims to resolve geographic name ambiguity. For instance, a street name like "State Street" exists in numerous American cities. In these cases, we integrate IP geolocation databases to provide auxiliary disambiguation support. Previous research [13, 16, 38] demonstrates that integrating multiple IP databases provides reliable city-level positioning. By analyzing the device's IP address location, the system constructs a regional constraint at the city or provincial/state level. This allows it to identify the most plausible geographic match among multiple candidates, significantly enhancing the final credibility of the inference results, particularly for identically named but geographically distinct locations.

5 Evaluation of framework

5.1 Datasets

5.1.1 Original Corpus. Our research utilizes the public Censys dataset [14], from which we filter 205,028 devices tagged as SCADA. This tag is selected because SCADA systems represent a typical type of remote management infrastructure, helping to refine our research scope and providing clear target indicators for our web page queries. We access their web interfaces in a strictly ethical and non-intrusive manner, comparable to how an ordinary user visits a public site via a browser. Any endpoint requiring authentication or that is

inaccessible is immediately abandoned, with no attempts made to bypass or attack. This approach naturally filters out protected devices, yielding a final corpus of 89,477 content-rich public pages for our analysis.

5.1.2 Clustering Threshold Tuning Dataset. To determine an optimal Hamming distance threshold for Nilsimsa-based clustering of remote management device interfaces, we manually identify 54 distinct page classes from the 89,477 pages corpus that potentially contain geographic information markers. These classes are selected specifically because they demonstrate consistent patterns of embedded geographic location data within their interface structures. Using these representative page templates, we identify and extract 10,709 structurally similar pages to construct a specialized device type dataset. This curated collection enables precise calibration of clustering similarity thresholds essential for accurate device categorization.

5.1.3 Gold Standard. For system training and evaluation purposes, we manually construct a comprehensive *gold standard* dataset. From the 10,709 web pages threshold tuning dataset, we methodically select pages containing identifiable geographic location indicators. This reference dataset incorporates multiple essential components: original HTML pages sourced directly from remote management devices, extracted differential text segments that distinguish individual devices, corresponding search engine query results derived from these differential text elements, and verified geographic location information extracted from these query results.

We implement a rigorous annotation protocol, assigning each candidate page to multiple volunteers and ensuring independent evaluation by at least two annotators per page. Only pages with complete inter-annotator agreement are retained in the final dataset. Out of ethical considerations, we do not make this data publicly available; however, our detailed data processing methods sufficiently support the data construction. This labor-intensive curation process ultimately yields a gold standard collection comprising 1,408 fully validated entries that serve as our definitive evaluation benchmark.

5.2 Experiment Setup

5.2.1 Web Page Differential Text Extraction. We utilize BeautifulSoup to parse HTML content, extracting text and DOM tree features which are concatenated to form a composite string for Nilsimsa hash generation. Similarity between pages is calculated using Hamming distance with a threshold of 40 (analysis in Section 5.6). Our clustering algorithm iterates through hash values, assigning each page to an existing cluster or creating a new one. The first page added determines each cluster's representative hash. For differential text extraction, we select one reference file per cluster and compare remaining files against it.

5.2.2 Geographical Clue Mining. In the geographic keyword matching stage, we employ a list of 65 keywords (Appendix Table 3) with case-insensitive matching to maximize identification accuracy.

In the search engine information augmentation stage, we implement programmatic control of Google Chrome using Selenium with ChromeDriver for automatic browser operation. Bing serves as our primary search engine due to its comprehensive result summaries,

related images, and regional prioritization features that enhance geographic information extraction. Bing's interface also demonstrates better resistance to anti-automation controls, making it suitable for batch query operations.

In the multi-model collaborative inference stage, We evaluate both open-source models (Mistral, Gemma2, GLM4, LLaMA3.1) and closed-source models (ChatGPT-4o, Claude 3.7) for our framework. We select Mistral-7b, GLM4-9b, Gemma2-9b, and LLaMA3.1-8b specifically for offline deployment compatibility with the Ollama tool, while accessing ChatGPT-4o and Claude 3.7 through online APIs. All models are evaluated using the *gold standard* dataset from Section 5.1.3.

In the geospatial disambiguation with database assistance stage, we integrate location data from IP2Location, IPinfo, MaxMind, and IPIP platforms. When discrepancies arise among database results, we apply majority voting to determine the definitive geographic location.

5.3 Evaluation Metrics

To comprehensively evaluate our method, we established evaluation metrics across two dimensions: the effectiveness of HTML page clustering and the capability of geographic information extraction.

5.3.1 Evaluation of HTML Page Clustering Effectiveness. We employ a suite of standard metrics to assess the performance of the clustering algorithm:

Purity: Evaluates the consistency of data point labels within each cluster.

Adjusted Rand Index (ARI) & Normalized Mutual Information (NMI): Both measure the similarity between the clustering results and the true class labels. ARI is corrected for chance, while NMI approaches it from an information-theoretic perspective.

Homogeneity, Completeness & V-measure: Homogeneity assesses whether each cluster contains only samples of a single class. Completeness assesses whether all samples of a given class are assigned to the same cluster. The V-measure is the harmonic mean of these two, providing a single, comprehensive score.

5.3.2 Evaluation of Geographic Information Extraction Capability. We use two core metrics to assess the effectiveness of the LLM in extracting geographic information:

Coverage: Measures the proportion of pages from which the model successfully extracts geographic information out of the total number of pages that actually contain it. This metric reflects the model's "recall capability."

Accuracy: Measures the proportion of correctly extracted geographic locations among all locations extracted by the model. This metric reflects the model's "precision."

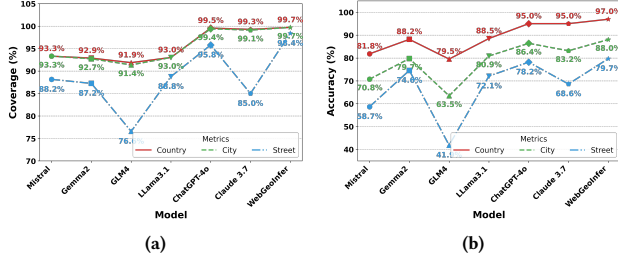
5.4 Overall Performance

To directly demonstrate the feasibility of our method in real-world, complex scenarios, we conducted an end-to-end performance evaluation of the entire framework on the previously constructed ground-truth dataset.

The experimental results show that our framework performs exceptionally well on this highly challenging dataset, achieving a city-level coverage of 99.71% and a street-level accuracy of 79.70%.

Table 1: Performance Comparison of Different Geolocation Methods

Evaluation Metric	Whois Query [15]	rDNS [10]	ProbeGeo [22]	Ours
Country-level Coverage	79.10%	6.80%	14.60%	99.71%
City-level Coverage	6.38%	6.78%	9.92%	99.71%
Street-level Coverage	—	—	9.78%	98.40%
Country-level Accuracy	51.95%	1.59%	3.05%	96.96%
City-level Accuracy	3.69%	0.28%	0.95%	88.05%
Street-level Accuracy	—	—	0.19%	79.70%

**Figure 3: Comparison of Large Language Model Performance. (a) Coverage and (b) Accuracy at Country, City and Street Levels.**

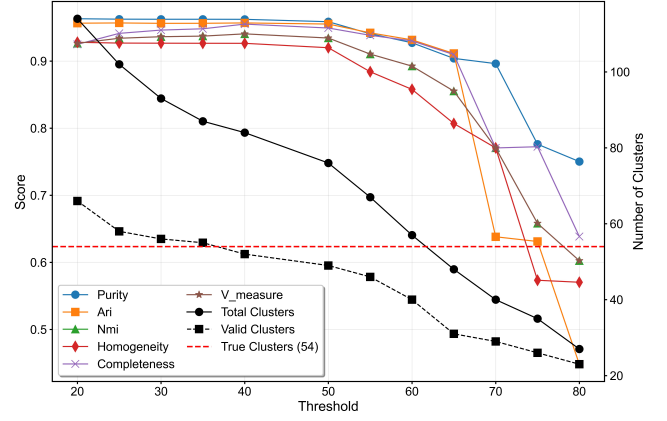
This success directly validates the effectiveness of our framework’s design.

To further verify the advantages of our framework, we also evaluated mainstream passive geolocation techniques on the same dataset, with the results shown in Table 1. These techniques include Whois lookups [15], rDNS analysis [10], and the ProbeGeo framework [22]. The results indicate that these methods are completely unable to handle the challenge: Whois achieved a city-level accuracy of only 3.69%; rDNS provided useful clues in just 6.80% of cases; and ProbeGeo (an NER-based method), which also relies on webpage content extraction, had a coverage of only 14.60%. The collective failure of these methods serves as reverse proof of the genuine difficulty of the problem we have defined.

In summary, the success of our framework on the ground-truth dataset indicates that our proposed approach, which combines information enhancement and deep reasoning, is the first highly viable solution to the challenge of extracting geographical locations at scale from non-standardized web pages. This reveals a previously overlooked potential security risk that warrants significant attention from regulatory bodies.

5.5 Comparison of LLM Performance

To evaluate the role of LLMs in our geolocation framework, we first conducted a preliminary test on 100 gold standard samples to assign performance-based weights: the top-performing online models, ChatGPT-4o and Claude 3.7, received a weight of 2; the best offline model, LLaMA 3.1, received a weight of 1.5; and other models were assigned a weight of 1. Subsequently, we conducted a comprehensive evaluation of all models for coverage and accuracy across country, city, and street granularities on the entire gold standard dataset.

**Figure 4: Clustering evaluation metrics under different thresholds. “Total Clusters” is the original cluster count, “Valid Clusters” is the effective count after noise removal, and “True Clusters” is the ground truth number of classes.**

As shown in Figure 3, there was significant performance variance among the models. The online models, ChatGPT-4o and Claude 3.7, performed best overall, with over 99% coverage at the country and city levels and a shared country-level accuracy of 95.03%. ChatGPT-4o particularly excelled at the most challenging street-level extraction, leading with 95.79% coverage and 78.20% accuracy. In contrast, GLM4’s street-level accuracy was only 41.87%, revealing its limitations in fine-grained extraction.

Ultimately, our ensemble method surpassed the best performance of any single model across all evaluation metrics. Notably, it achieved the most significant improvement at the street level—the most difficult task where individual models struggle most. This consistent improvement across all granularities strongly validates the effectiveness of our strategy of leveraging the complementary strengths of multiple models.

5.6 Hyperparameter Experiment

Analysis of Different Threshold Settings in Differential Text Extraction: We evaluate clustering performance across threshold values from 20 to 80 using the dataset constructed in Section 5.1.2. A threshold of 40 is ultimately chosen because it offers the best balance between homogeneity and completeness. At this threshold, 83 clusters are formed: 52 clusters (size ≥ 2) closely align with 54 reference categories, while the remaining 32 singleton clusters (accounting for approximately 3.1% of the total data) are treated as noise. The V-measure reaches its highest value of 0.94.

5.7 Ablation Experiment

5.7.1 Analysis of Clustered Differential Text Extraction Strategy. To demonstrate the effectiveness of our clustering algorithm in extracting differential page text, we directly input raw HTML into GPT-4o under the same experimental design and obtained coverage and accuracy rates at the country, city, and street levels. The results are shown in the Table 2. Our differential text extraction strategy significantly outperforms the use of raw HTML. This gap

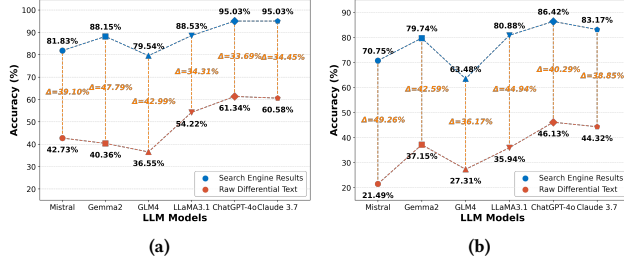


Figure 5: Accuracy Comparison of LLMs on Raw differential text vs. Search Engine Query Results. (a) Country-Level, (b) City-Level.

Table 2: Performance Comparison of Clustering-Based Differential Text Extraction Strategies and Raw HTML Usage

Evaluation Metric	Raw HTML [15]	Ours
Country-level Coverage	45.20%	99.71%
City-level Coverage	41.70%	99.71%
Street-level Coverage	26.60	98.40%
Country-level Accuracy	32.01%	96.96%
City-level Accuracy	13.03%	88.05%
Street-level Accuracy	4.96	79.70%

demonstrates that clustered differential extraction is essential for removing reasoning obstacles (manufacturer noise, abbreviated clues), not merely reducing tokens.

5.7.2 Search Engine Information Augmentation Analysis: To validate the effectiveness of information enhancement based on search engines within our framework, we design and implement ablation experiments. These experiments aim to compare performance under the following two scenarios:

- Directly identifying geographic location clues from the original differential text.
- Geographic location clue identification after augmenting the original differential text with search engine query results.

We evaluated performance differences between methods using country and city-level accuracy metrics (Fig.5a and Fig.5b). Our ablation experiments show that search engine enhancement significantly improved geographic inference performance, increasing country-level accuracy by 34.31%-47.79% and city-level accuracy by up to 49.26%. These results confirm the effectiveness of our search engine information enhancement approach.

5.7.3 Comparative Analysis of NER and multi-model collaborative inference: To verify the advantages of our multi-model collaborative inference approach compared to NER methods, we compared both approaches' coverage rate and accuracy under identical conditions, using search engine query results to enhance the original differential text. As shown in Figure 6, our method achieves superior performance in coverage rate and accuracy across different

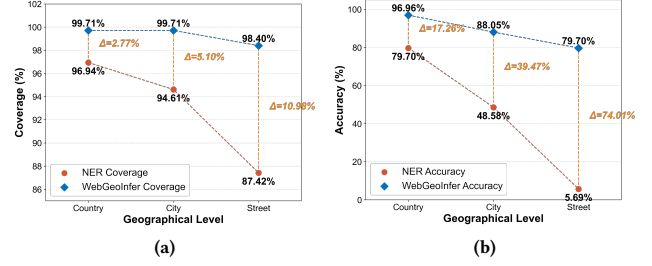


Figure 6: Comparison of NER vs. Multi-model Inference for Geographic Information Extraction (a) Coverage and (b) Accuracy at Different Geographic Levels.

levels, particularly in street-level accuracy, where it significantly outperforms NER methods.

6 Scope of Application and Results Analysis

We analyzed 89,477 accessible web pages for remotely managed SCADA devices and successfully inferred the geographic locations of 5,435 devices (2,305 via keyword matching and 3,130 via a large language model). We also analyzed the device brands and their distribution in cyberspace.

6.1 Device Brand Distribution

We conducted a brand analysis on the 5435 successfully located devices, identifying 33 device models across multiple critical infrastructure sectors (Appendix Table 4 for details). The diversity of device types, such as 967 Four-Faith power grid devices and 320 Obvius data acquisition devices, not only confirms the representativeness of our SCADA sample filtered from Censys but also validates our method's effectiveness in handling highly heterogeneous web pages.

More importantly, the analysis reveals the direct value of geolocation in security risk assessment. We found that a high proportion, 38.6% (2097 devices), use the "Niagara Web Launcher" component. Given that the Niagara Framework has multiple known vulnerabilities, the widespread use and geographic clustering of this component, as revealed by our geolocation (Appendix Figure 7), directly expose potential single-point-of-failure risks in the remote management supply chain and significantly expand the theoretical attack surface.

This finding demonstrates our core objective: geolocation inference is not only feasible but also a critical capability for identifying and assessing regional, large-scale infrastructure security risks.

6.2 Geographic Distribution Analysis

According to the statistical analysis of the results, (Appendix Figure 7 and Figure 8), the geographic distribution of remote management devices worldwide is extremely uneven. At the country level, the United States leads by a wide margin with 4,219 devices, and the US and Canada alone account for 80% of the total, forming a typical long-tail distribution. This high concentration reflects the

first-mover advantage of certain countries in industrial digitalization, but it also creates a strategic concentration of geopolitical and security risks.

Analysis at the city level further reveals that within individual countries, these devices also exhibit clear regional clustering, primarily in industrial centers, manufacturing hubs, and technology innovation zones. This pattern confirms the strong correlation between the deployment of digital infrastructure and a city's economic function and level of development.

6.3 AS Distribution Analysis

We successfully mapped the IP addresses of 3,954 out of 5,435 devices to their corresponding Autonomous Systems (ASes), finding they belonged to 336 distinct ASes. The distribution exhibits a significant long-tail characteristic (Appendix Figure 9): over 60% of the devices are concentrated within the top five largest ISP networks, while the rest are scattered across nearly 300 smaller ASes.

This distribution pattern presents a dual security challenge. On one hand, the high concentration of devices in major ISPs facilitates systematic scanning and large-scale attacks, creating a "centralized risk." On the other hand, numerous high-value devices are dispersed among smaller, specialized ISPs with inadequate security capabilities, forming "vulnerable edge points" and regulatory blind spots for critical infrastructure.

Furthermore, some organizations have a large number of exposed devices across multiple ASes they own (Appendix Figure 10), which directly indicates a systemic failure of their network boundary protection strategies and reveals a "broken chain of security responsibility." In summary, this situation reflects a disconnect between the internet's layered architecture and practical cybersecurity measures. The "default open" policies of large ISPs, combined with the lack of security resources in smaller ASes, have led to the uncontrolled exposure of managed devices on the public internet, highlighting the collective action problem in current distributed network governance.

7 Discussion

This section is focused on our core objective: demonstrating that automatic geographic inference from public web pages is critical for risk governance. We will briefly address the limitations, strongly affirm our ethical stance, and propose concrete security recommendations based on our findings.

7.1 Limitations and Research Value

Our method's effectiveness is contingent on the presence of geographic clues in web pages, and deploying LLMs at scale presents cost challenges. Future work could focus on developing a lightweight, specialized model for this task. However, these limitations do not diminish our research's core value: we have confirmed that a significant number of devices are already exposing sensitive information on the public internet to be automatically geolocated, a de facto situation that constitutes a major, pressing security warning.

7.2 Ethical Considerations and Declaration

We must emphasize that this study strictly adheres to ethical guidelines. Its sole purpose is to reveal risks through entirely non-invasive

methods, not to exploit vulnerabilities. We exclusively utilized the public Censys dataset, meticulously filtering entries pre-labeled as SCADA systems. Our process exclusively simulated a standard user's browser behavior: we only collected HTML pages from public IPs and ports, immediately discarding any endpoints that were closed, unresponsive, or required authentication. We did not conduct any active probing, vulnerability scanning, or attempts to bypass security mechanisms. The core objective of this research is to alert the security community, device manufacturers, and regulatory bodies to the risks of information exposure, aiming to promote proactive defense strategies rather than facilitating attackers. We are committed to maintaining the confidentiality of the detected information and will make every effort to notify the corresponding service providers.

7.3 Security Recommendations

Our research not only proves the feasibility of geolocation inference but also uncovers systemic risks through geographic and network topology analysis. Based on these key findings, we propose the following targeted security recommendations, designed to translate our analysis into actionable defense measures:

For Regulatory Bodies: country CERTs and critical infrastructure protection agencies should employ similar techniques to map their nation's exposure surface and assess regional cascading risks. Accountability and oversight mechanisms should be established for ISPs that host a disproportionate number of exposed devices.

For ISPs: Large ISPs should move away from "default-open" business practices and instead provide secure-by-default network configurations for enterprise clients, along with tools to help them audit their public-facing attack surface.

For Device Administrators: Security awareness training is crucial. During device configuration, avoid entering sensitive data like street addresses, contact numbers, or client names into publicly visible fields. Implement robust access control policies to ensure that only authorized personnel can access management interfaces containing location information.

8 Conclusion

This work confirms the significant threat of geolocation leakage from remote device web interfaces and proposes the WebGeoInfer framework. It overcomes information heterogeneity via clustering and differential analysis, while tackling information sparsity using search engines and LLMs. Our large-scale experiments located 5,435 devices across 94 countries and 2,056 cities, achieving high accuracies of 96.96% (country), 88.05% (city), and 79.70% (street). Beyond proving the threat's reality and scale, this research provides novel insights, discusses mitigations, and establishes a critical benchmark for future security and regulatory work.

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A Appendix

Algorithm 1 Differential Text Extraction

```

1: Input: Clusters  $C = \{C_1, \dots, C_m\}$  from Algorithm 2
2: Output: Differential text collection  $\Delta$  for each cluster
3: Initialize empty collection  $\Delta \leftarrow \emptyset$ 
4: for each cluster  $C_k \in C$  do
5:   Let  $C_k$  contain pages  $\{p_{k1}, p_{k2}, \dots, p_{kn}\}$ 
6:   Parse each page  $p_{ki}$  into a DOM tree  $T_{ki}$ 
7:   Identify isomorphic subtrees across  $T_{k1}, \dots, T_{kn}$  using sub-
   tree isomorphism
8:   Obtain aligned node groups  $G = \{g_1, \dots, g_r\}$ 
9:   Initialize differential text set  $\Delta_k \leftarrow \emptyset$ 
10:  for each aligned node group  $g_j \in G$  do
11:    Extract text contents  $\{text(g_j, p_{ki}) | i = 1, 2, \dots, n\}$ 
12:    Normalize text (remove extra spaces, unify case, etc.)
13:     $textSet \leftarrow \{text(g_j, p_{k1}), \dots, text(g_j, p_{kn})\}$ 
14:    if there exist different texts in  $textSet$  then
15:       $\Delta_k \leftarrow \Delta_k \cup textSet$ 
16:    end if
17:  end for
18:   $\Delta \leftarrow \Delta \cup \Delta_k$ 
19: end for
20: return  $\Delta$ 

```

Table 3: Keyword Categories for Deterministic Geographic Matching

Category	Specific Types
Common Road Types	street, avenue, road, boulevard, highway, lane, way, drive, path, expressway, parkway, alley
Venue and Business Types	market, shop, store, boutique, supermarket, pharmacy, bank, library, museum, restaurant, hotel, inn, resort, mall, plaza, center, square, park, garden, stadium, theater, cinema, arena, club
Administrative and Community	city_hall, courthouse, police_station, fire_station, community_center
Transportation Venues	airport, station, terminal, bus_stop, subway, metro
Special Types	bridge, castle, monument, landmark, district, neighborhood
Administrative Divisions	city, town, village, county, state, province, region, municipality
Country and Continental Levels	country, nation, continent, territory

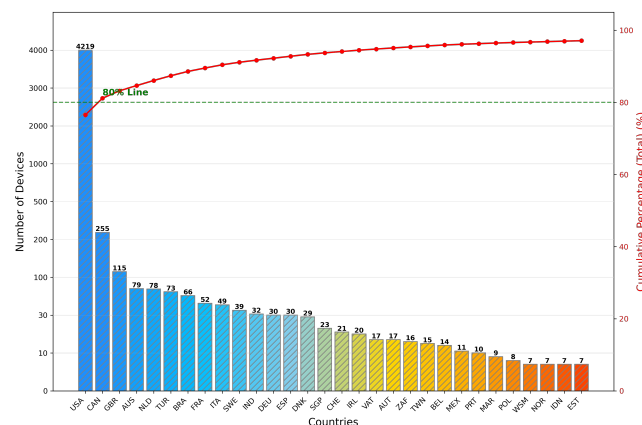


Figure 7: Country-level Distribution of Remote Device Management Inference Results.

Algorithm 2 Nilsimsa-based Device Page Clustering

```

1: Input: A set of pages  $\mathcal{P}$ , similarity threshold  $\theta$ 
2: Output: Clusters  $\mathcal{C}$ 
3: Initialize hash set  $\mathcal{H} \leftarrow \emptyset$ 
4: for each page  $p_i$  in  $\mathcal{P}$  do
5:   Extract DOM tree  $T_i$ , serialize to  $S_i$ 
6:   Compute Nilsimsa hash  $h_i \leftarrow \text{NilsimsaHash}(S_i)$ 
7:    $\mathcal{H} \leftarrow \mathcal{H} \cup \{h_i\}$ 
8: end for
9: Build similarity matrix  $M$ 
10: for each pair  $(h_i, h_j)$ ,  $i \neq j$  do
11:   Calculate distance  $d \leftarrow \text{HammingDistance}(h_i, h_j)$ 
12:    $M_{i,j} \leftarrow (128 - d)/128$ 
13: end for
14: Initialize single-page clusters  $\mathcal{C} \leftarrow \{\{p_1\}, \dots, \{p_n\}\}$ 
15: while there exist  $C_a \neq C_b$  s.t.  $\text{AvgSim}(C_a, C_b) \geq \theta$  do
16:   Select pair with maximum average similarity
17:   Merge clusters
18:   Update cluster collection  $\mathcal{C}$ 
19: end while
20: return  $\mathcal{C}$ 

```

Table 4: Distribution of Device Brands and Industries Exposing Geographic Location Information

Device Model/Brand/Component	Industry	Count
Niagara Web Launcher	Web Component	2097
Four-faith	Power Grid Automation	967
Carrier	Home Air Conditioning	520
Obvius	Data Acquisition Services	320
XFLOW (Schneider)	Industrial Remote Management	187
ASPECT (ABB)	Home & Building Automation	128
SIMATIC HMI	Human-Machine Interface	102
Dorian Systems	Building Thermal Control System	79
LOYTEC	Controllers	76
Kabona	Energy Automation	54
Ignition	Industrial Control Software	43
Mofi network	Network Management	43
Blue Iris	Video Surveillance	29
MOBOTIX	Video Surveillance System	24
CKSquare	Commercial Manufacturing	19
InteLift	Oil Well Management System	17
Perle	Network Management	9
Telkonet	Energy Management System	8
Perax	Remote Management System	6
Ipecs	Hybrid Communication Platform	6
Digi	Network Management	6
NetBiter	Remote Monitoring & Industrial Management	5
Delta Controls	Building Automation Control	5
TNT Technologies	Infrastructure	5
Rain Bird	Irrigation System Controller	4
Avaya	Network Systems	4
Dividia	Security Assurance	4
Coster Group	Building Energy Management	3
RC-WebView	Building Management	3
Trihedral	SCADA Software	3
Carlo Gavazzi	Photovoltaic Applications	3
AMAECO	Energy Management	3
PowerLogic (Schneider)	Power Management	3
Others	No obvious branding	650
Total		5435

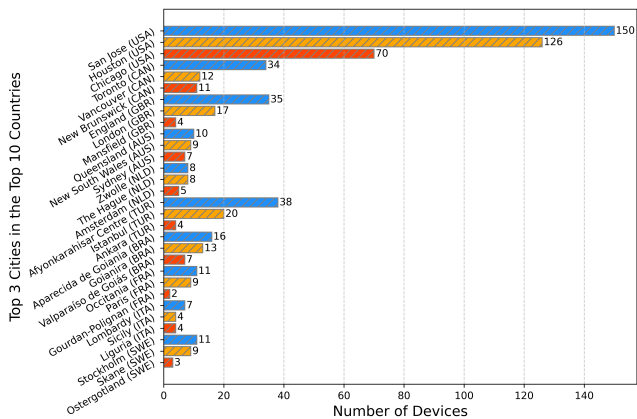


Figure 8: Top 3 Cities with the Most Devices in Top 10 Countries of Remote Device Management Inference Results.

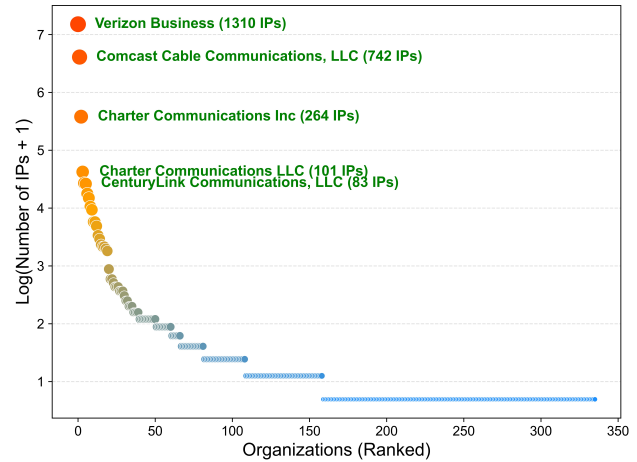


Figure 9: Distribution of Exposed Remote Management Devices Within Top 20 Organizations

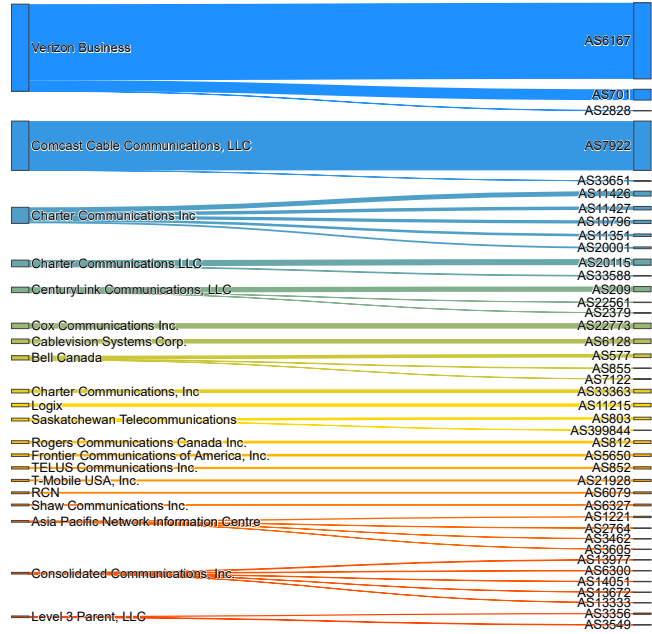


Figure 10: This Sankey diagram maps organizations (left) to Autonomous System (AS) domains (right), with line thickness representing relative device counts per AS to visualize distribution across network infrastructures.